

Task 9.5 – Linear and Logistic Regression from Scratch

Objective

The objective of this project was to implement Linear Regression and Logistic Regression algorithms from scratch using Gradient Descent and evaluate them on real-world datasets.

Datasets

Regression Dataset

- oil_sales_assignment_dataset.csv

Target:

- value_sales

Features included:

- year
- month
- volume_sales
- average_price

Classification Dataset

- heart_disease_risk_2026.csv

Target:

- has_heart_disease

Several numerical health indicators were used as input features.

Linear Regression

Implemented:

- Gradient Descent
- Prediction
- MSE
- RMSE
- R^2 Score

The trained model achieved a good fit on the regression dataset.

Logistic Regression

Implemented:

- Sigmoid Function
- Gradient Descent
- Binary Prediction
- Accuracy

The model successfully classified patients based on health indicators.

Output Files

Generated:

- linear_regression_predictions.csv
- logistic_regression_predictions.csv

Libraries Used

- NumPy
- Pandas

Conclusion

Both Linear Regression and Logistic Regression were successfully implemented from scratch without using machine learning libraries. The models were trained on real datasets, evaluated using appropriate metrics, and prediction files were generated for further analysis.